

FIFTH SEMESTER DIPLOMA EXAMINATION IN ENGINEERING/TECHNOLOGY

COMPUTER HARDWARE & NETWORKING

MODEL QUESTION PAPER

(EL)

(Time : 3 hours)

(Maximum marks:100)

PART – A

(Answer *all* questions in one or two sentences. Each question carries 2 marks)

I

1. Draw the pin layout of keyboard Mini-DIN (PS/2) connector?
2. Define 'form factor' of a motherboard.
3. Define the term 'Low level formatting'.
4. State the need for driver software.
5. Give any two examples of guided transmission media.

(5x2=10)

PART – B

(Answer any *five* of the following questions. Each question carries 6 marks)

II

1. State the output voltage levels with corresponding colour codes of an ATX SMPS
2. Explain the working principle of a digital camera.
3. Describe any three expansion slots in a motherboard.
4. State the need of CMOS chip in motherboard.
5. Compare FAT16 & FAT32
6. Describe 'Bus' & 'Ring' topology
7. Describe VPN (Virtual Private Network)

(5X6=30)

PART – C

(Answer *one* full question from each module. Each question carries 15 marks)

MODULE – I

III

- (a) Describe the method of encoding a key press in a key board. (7)
- (b) Explain the working principle of an inkjet printer with relevant diagram (8)

OR

IV

- (a) Describe the working principle of an ATX SMPS with block schematic (9)
- (b) Describe the working principle of an optical mouse (6)

MODULE – II

V

- (a) Analyse the role of BIOS and POST in a PC System (8)
- (b) Compare different types of Memory Packaging (7)

OR

VI

- (a) Describe the role of Graphics card in a PC (7)
- (b) Compare SRAM and DRAM (8)

MODULE – III

VII

- (a) Analyse the concept of free software (6)
- (b) With the help of diagram, describe the constructional details of hard disk (9)

OR

VIII

- (a) Describe the WINDOWS OS Installation steps (8)
- (b) Explain the principle of CD recording (7)

MODULE – IV

IX

- (a) Describe ISO-OSI reference model, with relevant diagram (9)
- (b) Write Short notes on (i) Wi-Fi (ii) Bluetooth (6)

OR

X

- (a) Describe the concept of e-mail & world wide web (8)
- (b) Compare Router and Bridge (7)

(4X15=60)